

Re: Strength of Property Rights and Private Ownership Subindex Evaluation

Dear Professor Bruner,

Thank you for your thorough and thoughtful evaluation of the Global Capitalism Index and the Strength of Property Rights and Private Ownership subindex. We greatly value your critiques, and we appreciate the care you took in reviewing multiple elements of the index. We have made substantive changes and additions to this subindex based on your recommendations and hope they sufficiently address your concerns. Where we made no changes, we provide a detailed explanation. Below are our responses to your feedback and recommendations.

I. General Comments

Audience

“As you prepare for a public launch of the indexes, consider preparing some messaging appropriate for each segment of the audience. The white paper serves as a technical appendix for researchers that addresses the fine points raised in the methodology. Practitioners and consumers would benefit from a discussion that addresses how to use and interpret the results. More about this in the following sections.”

We agree that communicating the GCI to a public audience will be crucial for a successful launch. We are currently developing complementary materials tailored to broader audiences that focus on interpretation, use cases, and practical insights. More specifically, we are creating a public website that visualizes our results and guides users on how to read the index, what comparisons are most meaningful, and what analyses can be drawn from it.

Beyond academic readers, the GCI is designed to serve multiple sets of audiences. The public website will offer open interactive access to scores, subindex breakdowns, and country profiles, oriented toward general readers, journalists, and educators. Topical reports applying the index to specific questions on institutional change, political and economic shifts, and comparative dynamics will serve policymaker and journalist audiences. Private sector applications, particularly in emerging markets and country risk, will address a distinct constituency of asset managers, advisory firms, and institutional investors. Each tier will be supported by a deliberate dissemination strategy, ranging from op-eds and longer form commentary in outlets such as the Chronicle of Higher Education and Harvard Business Review to directly engage with practitioner and policy communities.

Motivation

“In an article for the Journal of Economic Education, I discussed the usefulness of data analysis in building knowledge, skills, and wisdom. The white paper and other materials neglect the instructional potential of the indexes. I suggest inserting some discussion along these lines.”

Thank you for identifying this important element we had not previously addressed in the white paper. The framework you lay out in your JEE article makes it apparent that we need to make the case explicitly for the index’s pedagogical value rather than leaving it for readers to infer. We map the knowledge, skills, and judgment sections you reference onto the GCI below:

- **Knowledge:** A persistent obstacle to teaching about capitalism that students encounter "capitalism" as an abstract and contested category rather than as a system whose properties vary observably across countries and over time. The GCI converts the abstract into the measurable. Eight subindices give students a concrete vocabulary for the dimensions on which capitalist systems actually differ and a broad set of country observations give them the cross-sectional variation needed to see that "capitalism" is a system of institutions, not a single object.
- **Skills:** Because the GCI is built bottom-up from 275 datasets through a transparent two-stage PCA, students can engage it at multiple levels of analytical commitment. At an entry level, they can download subindex scores, plot them against outcomes (growth, inequality, financial development), and formulate and test hypotheses. At an intermediate level they can examine the indicators inside a subindex, ask whether the aggregation choices we made are the right ones, and re-aggregate under alternative weightings. At an advanced level, they can replicate the construction from raw inputs using provided replication files, propose new indicators, or extend the panel. Each level exercises data development and curation, presentation, hypothesis testing, and synthesis across disciplines.
- **Judgment:** Every methodological choice in the index is a normative decision at some level. Everything from the dimensions that count as constitutive of "capitalism," to whether to treat institutional quality as separable from economic outcomes, to how to handle missing data for countries whose institutional opacity is itself informative. We have tried to be as transparent as possible about these choices in the construction. That transparency turns the index into an instrument students can argue with. A high-engagement classroom exercise built around "should the GCI weight subindex X more heavily?" or "what elements actually matter in this subindex" develops the judgment you describe.

We will add a dedicated section to the whitepaper detailing the instructional applications of the index and developing this argument further. Additionally, we will build a Teaching Resources area on the website with a set of structured exercises spanning the three levels above, designed for adoption in courses on comparative political economy, development, and finance. We are grateful for the push here and believe the pedagogical framing is a necessary component of the paper.

"So much of the argumentation for and against capitalism has to do with its association with outcomes about income and wealth, health, education, unemployment, equity, systemic stability, innovation, and governance (democracy vs. autocracy). Arguably, this improved index will strengthen research and the resulting claims about criticism and advocacy. The index will also make it easier for students to develop their own insights by exploring the association between capitalism and outcomes."

A core motivation for the GCI is to decouple the measurement of capitalist institutions from the outcomes they are often assumed to produce. By providing a rigorous measure of institutional inputs, we hope to enable researchers and students alike to empirically test claims about capitalism's relationship to "output metrics," including inequality, growth, innovation, and governance.

Principal Components Analysis

"It makes sense to use this technique for identifying the factors that explain the largest variation in the data. PCA is useful not only for identifying central tendencies, but also for detecting outliers (something that historians care about). Yet I would advise caution in how you present the methodology. First, the derived components can be

difficult to interpret (or even name) because each is an amalgam of original variables. Second, since the methodology assumes linear relationships among variables, I worry about (1) nonlinearities such as threshold effects due to critical tipping points, (2) jump functions due to external shocks or structural shifts, (3) diminishing returns to scale, and (4) heteroskedasticity (a bigger problem in financial data) all of which linear models will not fit successfully. And finally, given the number of components feeding into the subindex and global index, you might face the problem of overfitting that could limit the ability of the model to generalize well. The white paper gives a good technical discussion of PCA, though I wish it had more to say about items 1-4 and gave more argument about why the linearity of PCA gives a reasonable starting point for analysis.”

We appreciate this careful engagement with our methodology. We have added a discussion of these points to the methodology section starting on Page 32 and thank you for pushing us to be more explicit about the assumptions of our approach. Each area is addressed below.

On interpretability: We acknowledge that principal components can be difficult to interpret when factors are combinations of input variables. Our first-stage PCA extracts factors at the subindex level where the data together measure a clear conceptual domain. The second-stage PCA aggregates the eight subindices and creates an overall capitalism score based on the underlying variation in our subindices. We expand this discussion in the whitepaper’s “intepretation of component scores” section.

On nonlinearities: The linearity assumption in PCA pertains to the correlational structure among indicators within a subindex, not to the relationship between the GCI and any dependent variable. Nonlinear relationships between capitalism and outcomes are precisely the kinds of questions we hope researchers will investigate using the GCI as an independent variable. In a small handful of cases where we have strong theoretical reason to expect nonlinear measurement relationships, we apply parabolic transformations before the data enters the PCA process. For example, government spending as a share of GDP is transformed so that extreme values in either direction receive lower scores, reflecting the logic that both too little and too much state expenditure can undermine market functioning. This forms a parabolic curve with a predetermined “peak” value that yields a score of 10. We determine that peak by taking the average of a set of country values across that we feel cover the range of “types” of capitalism. We document these countries and the process further in the whitepaper.

On structural shifts and external shocks: Our period-specific estimation approach addresses this directly. Rather than estimating a single set of factor loadings across the entire 2008–2024 panel, we allow the correlational structure to shift between periods, with the data harmonization process ensuring cross-period comparability. If the 2008 financial crisis or COVID altered the relationships among institutional indicators, the model captures that structural change rather than forcing a single linear mapping across fundamentally different regimes.

On heteroskedasticity: Our winsorization procedure at the 1st and 99th percentiles reduces the influence of extreme observations that are often the primary drivers of heteroskedastic variance in cross-national data. The bootstrap procedure provides confidence intervals that reflect the variability in the estimates.

On overfitting: Our minimum factor loading threshold of 0.3 excludes variables that do not contribute meaningfully to extracted components. Our bootstrap procedure (1,000–2,000 iterations) generates confidence intervals that provide a direct measure of estimation uncertainty. If overfitting is producing unstable estimates, this appears as a wider confidence interval.

Promotion, Adoption, Impact

“How will you measure the impact of the index project? Impact will depend on adoption of the indexes by researchers, practitioners, and consumers. Rough measures of adoption would be number of downloads by kind of audience, number of research papers using the indexes, website “hits,” number of cross-listings to other websites, etc. And adoptions follow from an effort to promote the indexes as research tools. How will you promote the indexes?”

We plan to track adoption through several channels: downloads of the dataset and white paper, academic citations, media mentions and references in policy documents, and interactions on our web platform. We will also track usage by audience type where possible, distinguishing between academic, policy, journalistic, and private sector engagement. We also hope to track user results by country, thus allowing us providing first-hand data on how users around the world see and interpret capitalism.

Our promotion strategy operates on three levels. First, we plan to submit methodological and analytical academic papers to peer-reviewed journals, present at major conferences in political economy, comparative politics, and finance, and engage directly with scholars working on related indices and institutional measurement. Second, we are currently developing the interactive web platform that will allow journalists, policymakers, students, and the general public to explore country scores, compare subindex profiles, and download all data. Third, we plan to explore financial applications of the GCI that are relevant to investment professionals, corporate strategy teams, and policy practitioners. Finally, we are already planning a blitz of op-eds, podcast appearances, and media exposure to introduce the GCI to potential users across the sectoral spectrum.

Definition of Capitalism

“I see the importance of the institutional perspective (and agree with it) to distinguish this index from Heritage and Fraser, who take a more libertarian view. But I wish your definition had more grounding in “market activity” and open market prices as the defining features of capitalism. Branko Milanovic argued in a book that China is a capitalist country. Surely, China and North Korea can claim to have plenty of institutions for deploying capital and managing market activity, but does that make them capitalist? The U.S., with its layers of subsidies, tax breaks, regulations, and mandates is at best a “mixed economy”—would the U.S. qualify as capitalist under your definition? In an earlier blog post (see <https://blogs.darden.virginia.edu/brunerblog/2024/10/what-is-capitalism-why-definitions-matter/>) I argued for a definition based on voluntary exchange and open-market prices as the key attributes of capitalism.”

We share your emphasis on voluntary exchange and open-market pricing as vital observable features of a functioning capitalist system, but we believe that they do not emerge or sustain themselves without institutional preconditions. Voluntary exchange requires enforceable property rights and contracts. Open-market prices require competitive market structures, transparent regulatory environments, and functioning capital markets. The GCI’s institutional approach measures the systemic conditions that make voluntary exchange and open pricing possible rather than measuring the exchange and pricing activity itself.

We aim to demonstrate that systemic institutions that support outputs such as voluntary exchange and open-market pricing, *most often and in most circumstances*, make a capitalist economy. At

least in theory (though very rare in practice), exogenous circumstances arise that would allow for occurrences of voluntary exchange and open-market pricing in distinctly non-capitalist economies. We argue that those occurrences would be the exceptions—caused by exogenous events rather than true movements toward capitalism—that prove the rule. Put another way, in January 2018 it snowed for nearly 4 days in significant parts of northwestern Algeria. The storm eventually left nearly 2 feet of accumulation. Of course, no one suggested that the storm meant that northwestern Algeria was not, or ceasing to be, a desert climate. To make this argument, scientists would need to identify significant changes in the Hadley Cell, shifts in the nearby ocean currents, alterations to topographic blocks, and disruptions of low humidity feedback loops. We believe that the evaluation of capitalist systems should emphasize similar structural forces rather than the outputs that do and can be subject to rare co-incidents of exogenous forces. A country that receives a high weighted average score across all eight subindices, and thus receives a high capitalism score, will be *by definition one* in which voluntary exchange and market prices are highly likely to occur on a consistent basis.

This layered subindex design allows us to do a more robust and nuanced evaluation of analyses like Milanovic’s view of China as capitalist. His argument rests on the observation that China has extensive market activity, private capital accumulation, and wage labor. GCI results show that China’s market activity occurs within an institutional environment that is strong on some subindices and weak on many others. It empirically distinguishes China’s institutional configuration from both liberal market economies and non-capitalist systems. *How* it is distinguished will be one of GCI’s most important interventions. We will be developing a sophisticated typology of market modalities that goes beyond simple characterization of “state involvement” (what is the state doing? To what degree? What is the shape of the curve of state activity?) or “mixed economy” (Mixed in what way? What are the shapes of the distribution curves? What are the combinations of the components that are “mixed”?).¹ We hope this will add analytical rigor to questions about what types of systems qualify as “capitalist.”

II. Detailed Comments

Concentration of Datasets

“V-Dem data series account for six of eight series for “Lack of Corruption,” and five of seven for “Rule of Law.” I would feel more comfortable if the data sources for these two components were more broadly diversified. In other sub-indexes, I also note concentrations (e.g. in the Market-Supporting Policy index, of the 22 data series, 19 are drawn from WEF).”

This is an appropriate concern and one we have taken steps to mitigate. For Lack of Corruption and Rule of Law, the concentration in V-Dem reflects the fact that V-Dem provides the most granular and methodologically transparent institutional assessments currently available at the coverage levels we require. We supplement V-Dem with World Governance Indicators, which draw on different underlying source surveys and expert panels than V-Dem, as well as the Corruption Perceptions Index and the World Justice Project’s Rule of Law Index.

¹We believe these deep and nuanced investigations are vital to gaining a true characterization of capitalist systems. Regarding state involvement, capitalism demands widely varying levels of state activity depending on the concept being measured. For example, highly capitalist systems require robust state enforcement of private property rights and support for competitive market systems. However, truly capitalist systems require little to no state ownership of the means of production or participation in actual market activity.

We would note, however, that source concentration is at least partially mitigated by the fact that V-Dem’s individual indicators are derived from independent expert codings with documented inter-coder reliability, meaning that the six V-Dem series in Lack of Corruption are not six expressions of a single assessment but six independently coded dimensions with distinct expert panels. V-Dem’s world-class methodological precision and transparency gives us confidence that though we do concentrate certain elements of our subindex on their data series’, those series provide sufficient internal diversification so as to not bias our results.

Also, it is important to note that our PCA process ignores the walls between components. Our component categories provide a framework that ensures sufficient theoretical coverage for the concepts relevant to the subindex, but we conduct a single PCA across the aggregate collection of data within each subindex.

The Market-Supporting Policy subindex that you review was over-concentrated in WEF surveys and both reviewers noted the same issue. Thus, we have subsequently made significant changes to the composition of this subindex that added considerable diversity to the data sources. We will provide those updates to you with this response.

The Seven Components

“The selection of the seven topical components for property rights and private ownership seem appropriate, though they also seem likely to overlap. For instance, rule of law, lack of corruption, and contract enforcement would seem likely to yield similar assessments for the various countries. Can you sharpen the rationale for distinguishing these three from each other? Presumably the PCA methodology corrects for multicollinearity among these components. Also, I note that the five most important variables all cluster within just three of the components (see PowerPoint pages 6 and 7) does this imply that the other four components do not add meaningfully to the subindex?”

We view rule of law, corruption, and contract enforcement as related but meaningfully distinct institutional dimensions. Rule of law measures the extent to which agents have confidence in and abide by the rules of society. Lack of corruption measures the degree to which public power is exercised for private gain. Contract enforcement measures the practical capacity to resolve commercial disputes through legal mechanisms. A country can have strong formal rule of law but endemic corruption or low corruption but weak contract enforcement capacity. These are empirically distinguishable conditions with different implications for market functioning.

That said, we fully expect these components to be positively correlated, which is precisely what PCA is designed to handle. Since our PCA process does not assume orthogonality among input variables, it captures the common variance structure across correlated inputs and produces components that capture shared variation while downweighting redundancy. If rule of law, corruption, and contract enforcement were perfectly collinear, PCA would effectively treat them as a single signal. To the extent that they are correlated but not identical, the methodology extracts both their shared institutional quality dimension and any independent variation each contributes. Our oblimin rotation allows the extracted factors to be correlated, further accommodating the reality that institutional dimensions are interrelated rather than independent.

The slide deck presentation of the most important variables was misleading and apologize for the limited information presented. Despite that collection of datasets contributing more to the extracted factor, the others do contribute important additional variance that would be lost if they

were excluded. Additionally, the relative importance of components can shift across periods in our period-specific estimation, meaning that a component with lower average loadings may contribute more substantially in another period. We have added a discussion of loading distributions to the whitepaper to provide greater transparency on the relative contribution of each.

Market-Supporting Policy

“It is hard for me to tell from the titles of the data series, but the titles seem to imply that government intervention in the private sector is beneficial. In emerging economies especially, such intervention is a mixed blessing (see work by Hernando de Soto and William Easterley). I would be interested to see measures of the size of intervention relative to some basis such as GDP, industrial production, etc. Can you not find measures of state intervention by extent of business regulation, of state ownership in the private sector, of antitrust enforcement, and of incentives by the state in the form of cash subsidies (e.g., agricultural support), favorable (below market) credit terms, monopolies granted, or tax breaks? Some possibly useful measures might be: a) World Bank time required to obtain an operating license, World Bank subsidies and other transfers, OECD MAGIC database.”

The Market-Supporting Policy subindex is not designed to measure the quantity of government intervention or to assume that more intervention is inherently beneficial. Rather, it measures the quality and orientation of government policy toward creating and sustaining market systems. We believe that markets do not emerge or persist spontaneously but require active institutional maintenance from public and private actors. The subindex captures whether government policy is oriented toward creating the conditions for market activity, not whether government is large or interventionist.

State Stability Index

“You might consider sourcing estimates of stability from more sources especially private analysts, who have some economic “skin in the game”. See, for instance: World population review political risk index, Willis political risk index, PRS Group. Of special relevance to measures of state stability would be estimates of the political risk premium on the return required by investors across countries, and the default risk premium on bonds issued by national governments. Campbell Harvey (Duke) and Geert Bekaert (Columbia) have written several papers relevant to country risk. Some measures would include these: Kroll country risk measures (international cost of capital), Aswath Damodaran Data set at NYU.”

We have reviewed several of the suggested sources, including the PRS Group’s International Country Risk Guide, WTW index, Kroll’s country risk measures, and Damodaran’s country risk premium estimates. Regarding the PRS Group, Kroll, and WTS indices, these are proprietary commercial products that would require paid subscriptions to access and could not be redistributed alongside our published data. A core commitment of the GCI is that all underlying data sources be publicly accessible to the public once we publish results. Incorporating proprietary datasets would undermine this principle.

Regarding Damodaran’s country risk premium data, which is publicly available and which we do use in the Capital Market Sophistication subindex, we decided not to incorporate it in the state stability component because Damodaran’s risk premiums are derived from sovereign spreads and

CDS pricing. In Capital Market Sophistication, this data captures the functionality of a country’s pricing infrastructure. In state stability, however, it would amount to using a market outcome as a proxy for the institutional conditions that produce it, introducing circularity.

That said, we agree that additional measures of political constraints and institutional stability would strengthen this component. We added the POLCON index developed by Witold Henisz at Wharton to the state stability component. POLCON measures the constraints on policy change based on the structure of a country’s political institutions, namely the number of independent veto points and the distribution of preferences across them. It is publicly available, has broad country and temporal coverage, and captures the strength of structural constraints on arbitrary policy reversal.

Labor Market Openness Index

“It would be useful to reflect measures of the exercise of labor rights. Under labor protections, could you add a measure of days lost to strikes or other labor actions?”

We considered measures of labor action activity, including ILO’s days not worked due to strikes per 1000 workers. However, we decided not to include this measure because of interpretive ambiguity: a high number of strike days could indicate that workers possess and are able to exercise meaningful labor rights, which would suggest a well-functioning institutional environment, or it could indicate that labor relations are dysfunctional and bargaining mechanisms have broken down, which would suggest institutional failure. Conversely, a low number of strike days could reflect either effective dispute resolution and cooperative labor relations or the suppression of labor rights by the state or employers. We believe that days lost to strikes is an outcome of the labor relations environment, not a feature of the institutional infrastructure itself. We will note this consideration in the white paper discussion of the Labor Market Openness subindex.

Capital Market Sophistication Index

“When I hear experts discuss the leadership of the U.S. in international finance, they almost always cite the breadth and depth of the U.S. capital markets. It is not clear from the measures listed under this subindex that they capture this notion. For instance, can you include a measure of the percent of global capital market trading activity claimed by each country?”

The Capital Market Sophistication subindex includes measures of market depth, but we acknowledge that these indicators capture domestic depth relative to a country’s own economy rather than a country’s share of global capital market activity. The U.S. scores well on domestic depth measures but the fact that a disproportionate share of international transactions are denominated in dollars and cleared through U.S. markets is not directly captured. We considered measures of global trading share but ultimately did not include them because the GCI is designed to measure the institutional environment for capitalism within a country rather than a country’s relative position in the global financial system. A country’s share of global trading activity is partly a function of institutional quality but also reflects reserve currency status and network effects that are better understood as outcomes of institutional strength rather than inputs to it. Including such a measure would risk conflating the institutional infrastructure that enables deep capital markets with the consequences of having them.

Ordinal vs. Interval vs. Categorical Data

“I understand the requirement to convert the many different data series to a common (interval) scale. But a close reading of some of the data series invites doubts about such conversion. Page 17 of the PowerPoint cites the inclusion of a banking crisis dummy, and page 11 of the subindex memo states that the ”Expert survey question [is] posed as “Does the state own or directly control important sectors of the economy?” These two measures involve a categorical response rather than an interval (scaled) response. How does the transformation of data into a common scale deal with categorical data? Is this a pervasive issue in the various data series?”

We appreciate your close reading of the underlying data and address each point in turn. We have removed the banking crisis dummy variable from the Banking System subindex in the current iteration. Upon further review, we agreed that a binary indicator is not well suited to a PCA framework that assumes continuous variation, and we can better reflect the information it captures through the continuous indicators of banking stability already included in the component. Regarding the state ownership variable, we want to clarify that this is not a binary categorical response. The underlying question is answered with an ordinal response that is converted to interval data by a bayesian item response theory measurement model. It is measured on a continuous scale from 0 to 4, capturing gradations in the extent of state ownership rather than a simple yes/no determination. It enters the PCA as an interval-scaled variable consistent with the other indicators in the component. The specifics of the scale are detailed below:

- 0: Virtually all valuable capital belongs to the state or is directly controlled by the state. Private property may be officially prohibited.
- 1: Most valuable capital either belongs to the state or is directly controlled by the state.
- 2: Many sectors of the economy either belong to the state or are directly controlled by the state, but others remain relatively free of direct state control.
- 3: Some valuable capital either belongs to the state or is directly controlled by the state, but most remains free of direct state control.
- 4: Very little valuable capital belongs to the state or is directly controlled by the state.

We believe that the broader question of ordinal versus interval data is vital and we will address it in the white paper by detailing how we handle data differences. The majority of the GCI’s datasets are either continuous or interval-scaled. We utilize expert surveys and institutional assessments that are technically ordinal datasets presented as categorical data with 5+ categories (e.g., Likert-type scales from the WEF Executive Opinion Survey, or categorical ratings from V-Dem). The use of PCA with differing data types is common practice in index construction and is supported by a substantial methodological literature. Labovitz (1970) demonstrated that treating ordinal variables as interval-scaled and applying parametric techniques produces negligible error, particularly when scales contain five or more response categories.² Kolenikov and Angeles (2004) systematically examined the use of discrete data in PCA for socioeconomic index construction and found that PCA applied to ordinal variables with five or more categories produces results that closely approximate those obtained from more technically demanding approaches such as polychoric correlation matrix

²Labovitz, S. (1970). ”The Assignment of Numbers to Rank Order Categories.” American Sociological Review, 35(3), 515–524.

ces.³ Jolliffe and Cadima (2016) note that while PCA was developed under normality assumptions, as a descriptive tool it requires no distributional assumptions and has been successfully adapted for binary, ordinal, and discrete data across disciplines.⁴ We have added a discussion of variable type distribution across the GCI's datasets to the methodology section in the whitepaper. We have added a discussion of variable type distribution across the GCI's datasets to the methodology section in the whitepaper.

Interpretation and Significance

"The broader issue in the previous point deals with the interpretation of the index estimates: page 26 of the PowerPoint deck gives the comparison of average index results (2008-2023) for the U.S. (global index 74.64, rank 10) with nine other countries having index scores in the 70s (ranks 5 to 15). But these index values seem so close together as to render the differences in ranking possibly meaningless. Indeed, the Global Capitalism Index datasheet shows the confidence interval upper and lower bounds of the US and New Zealand overlap, as do other country pairs in the zone of ranks 5-15. The uncertainty around index values makes it less meaningful to declare that a country has a particular rank and instead invites an acknowledgement that a country falls into a certain group, such as a decile of all countries. What matters is the statistical significance of the difference in index values among countries. Presumably, the confidence interval upper and lower bounds for each country would afford a way to test for the significance of index differences among countries or groups of countries. It would be interesting to see a table of pairwise significance test statistics for all pairs of countries in the sample."

We agree that ordinal rankings can overstate the precision of the index, particularly in the middle and upper ranges where countries cluster tightly. Where confidence intervals overlap, a strict rank ordering implies a degree of differentiation that the data do not support. We have taken several steps to correct this issue.

First, we have added language to the paper emphasizing that GCI scores should be interpreted as estimates with associated uncertainty rather than as precise point values, and that rankings should be understood as approximate rather than definitive. Second, we present all country scores alongside their bootstrapped confidence intervals in the published data, enabling users to assess directly whether any two countries are statistically distinguishable. We will also include guidance in the methodology section on how to conduct pairwise comparisons using the published confidence intervals for researchers who wish to test specific country pairs. Third, we intend to include grouping analyses in the forthcoming results section where countries are clustered into tiers where within-tier differences are not statistically significant.

Missing Data

"The documents briefly mention an adjustment for missing data but does not explain the adjustment. Looking at the property rights data sheet, missing data is an endemic problem. Researchers will want to know more detail about treatment of missing data."

³Kolenikov, S. and Angeles, G. (2004). "The Use of Discrete Data in PCA: Theory, Simulations, and Applications to Socioeconomic Indices." Carolina Population Center, University of North Carolina, Working Paper 04-85.

⁴Jolliffe, I.T. and Cadima, J. (2016). "Principal Component Analysis: A Review and Recent Developments." Philosophical Transactions of the Royal Society A, 374(2065).

We addressed missing data through a sequential, three-stage procedure applied within each analytical period prior to factor scoring. In the first stage, observed values were smoothed along each country’s time series using a linear trend decomposition (trend-smoothing parameter = 0.5), which blends the raw observed value with its fitted trend. Missing observations are preserved as missing at this stage. The smoothing operates only on observed data points, avoiding the propagation of imputed values into the trend component.

In the second stage, remaining short gaps (up to three consecutive years) within each country’s time series were filled using linear interpolation. This approach is appropriate for panel data of this type, as it respects the temporal trajectory of each country rather than pulling values toward a cross-sectional mean.

In the third stage, values still missing after interpolation, primarily structural gaps where a country has no observations for a variable within a given period, were imputed using Predictive Mean Matching (PMM) via the MICE framework. PMM draws each imputed value from a pool of observed values with similar predicted scores, making no distributional assumptions and preserving the empirical range and variance of the data. We used a single imputation chain ($m = 1$) since uncertainty is separately quantified through bootstrap resampling ($n = 1,500$ iterations per period). Running multiple MICE chains within each bootstrap iteration would be computationally prohibitive and would not materially improve uncertainty estimates at that resampling scale. Within each bootstrap iteration, remaining missing values were handled via column-mean imputation, as sampling variance dominates imputation bias at $n = 1,500$. Variables with greater than 70% missing observations within a given period were excluded from the analysis entirely.

We have added our description of this missing data process to the white paper.

Regions and Other Aggregates of Countries

“Given the existence of zones of economic mutual interest (such as the European Union, COMESA, NAFTA, APAC, and Mercosur) and geographic neighborhoods (such as Scandinavia, the Middle East, and the former states of the USSR in central Asia), it would interest your various audiences to see index values for such aggregates.”

We agree that presenting geographical groups would enrich the analysis. We will include regional breakdowns as well as income groups in the forthcoming analysis section and on our interactive web platform.

Editing

“Page 41, paragraphs 1 and 2 of the white paper appear to duplicate a sentence. Also, in some places, the white paper cites 178 countries in the global sample, and in others it cites 175 countries.”

Thank you for catching those errors. We have removed the duplicated sentence on page 41 and standardized the country count throughout the paper. The correct preliminary figure was 175 countries, but we are in the process of finalizing that number.

Concluding Remarks

We are grateful for the rigor you brought to this review and the important questions raised across multiple dimensions of the index. In response to your feedback, we have made the following

substantive changes:

- 1: Expanded the methodology section to address PCA's assumptions, the treatment of ordinal and categorical data, and overfitting safeguards
- 2: Removed the banking crisis dummy variable from the Banking System subindex
- 3: Added the POLCON index to the State Stability component
- 4: Added language clarifying that GCI scores should be interpreted as estimates with associated uncertainty and that rankings reflect approximate groupings rather than definitive ordinal positions; committed to country groupings and regional aggregates in the forthcoming analysis section and public website
- 5: Added a loading distribution discussion to the appendix.
- 6: We have also corrected the editorial errors you identified.

Where we did not make changes, primarily regarding private data sources, global capital market share data, and days lost to strike, we have provided detailed explanations above. We hope these explanations adequately convey the reasoning behind our design choices.

We would be happy to discuss any of these points further, and we hope to continue this productive dialogue about how to strengthen the Strength of Property Rights and Private Ownership subindex and the Global Capitalism Index more broadly.

Sincerely,

UVA Democracy and Capitalism Lab